

General instruction(s):

Attend all the five questions

Q.No.	Question
1.	a. Give the specifications of IAS machine. (4 Marks) b. Write an Assembly code using IAS instructions for performing the following operations: $Z = X / Y$ Assume that inputs are available in the memory locations 501 and 502. Store the result in memory locations 503 and 504. (6 Marks)
2.	a. Draw the flowchart for the Modified Booth algorithm. (4 Marks) b. Perform multiplication of given two numbers, 15×-6 using Modified Booth algorithm. Description of each step is expected. (6 Marks)
3.	a. Discuss various representations of fixed point binary numbers with an example. (4 Marks) b. Apply the restoring division algorithm for the given expression. Description of each step is expected. (6 Marks) $14/6$
4.	a. Discuss about various representations of nonnumeric data. (4 Marks) b. Perform multiplication of the numbers -14 and 5 using Booth multiplication algorithm. Description of each step is expected. (6 Marks)
5.	Differentiate the computer architecture and computer organization and discuss the various functional components of a computer and the interaction between those components. (10 Marks)

General instruction(s):

Answer all five questions

Q.No.	Question	Max Marks
1.	List out the various registers of IAS machine and write the significance of each register. Also demonstrate an assembly code using IAS instructions for the following expression: $C = (A+B)/D$ Assume the inputs are available in the memory locations 101 onwards. Store the result in memory location 104 onwards.	10 [5+5]
2.	a. Draw the flowchart for Booth's algorithm. b. Perform Booth's multiplication for the given numbers -15 and 6 (i.e. -15×6). Description of each step is expected.	10 [4+6]
3.	a. Explain the IEEE 754 floating-point representation. b. Perform the division for the given numbers 21 and 4 (i.e. $21/4$.) using non-restoring algorithm. Description of each step is expected.	10 [4+6]
4.	a. Discuss the instruction types of IAS machine with suitable example? b. Describe the functional components of a computer with a neat diagram?	10 [5+5]
5.	Write the assembly code for the given arithmetic statement. $A = (B+C) * D$ using 0,1,2 and 3 address instructions and compute memory traffic for 0,1,2 and 3 address machines. Assuming the following fields: addresses - 2 bytes, data values - 2 bytes, opcode - 8-bits and word length - 1 byte.	10 [5+5]



Sec. 3

(a) Explain the internal structure of an IAS computer using a neat diagram.

(b) Suppose for an implementation of a **RISC ISA** there are 4 instruction types, with their frequency of occurrence and CPI as shown

5

1.

Type	Frequency	CPI
LOAD	20%	4
STORE	8%	3
ALU	60%	1
BRANCH	12%	2

Calculate average CPI.

2. (a) Tabulate the difference between RISC and CISC Architectures.

5

(b) Draw the block diagram of the Harvard architecture and explain why is Harvard architecture faster than Von Neumann?

5

Page 1 of 2

(a) Draw the flow chart of booth's algorithm.

(b) Show the steps of multiplication performed by using booth's algorithm of $(-12) \times (4)$

10

3. (c) Draw the flow chart of restoring division algorithm

(d) Divide the following binary format positive numbers using the restoring division algorithm: $A = 10101$, $B = 000011$, and verify your answer by converting A and B and your result (A / B) to decimal

- (a) Draw the flow chart of booth's algorithm.
- (b) Show the steps of multiplication performed by using booth's algorithm of $(-12) \times (4)$
3. (c) Draw the flow chart of restoring division algorithm 10
- (d) Divide the following binary format positive numbers using the restoring division algorithm: $A = 10101$, $B = 000011$, and verify your answer by converting A and B and your result (A / B) to decimal.
4. What are the advantages of hardwired control unit? Describe the design of hardwired control unit of CPU, with diagram. 10
- Let's consider three different processors P1, P2 and P3 and they execute the same instruction set. P1 processor has 3GHz clock rate and a CPI (Cycles Per Instructions) of 1.5, P2 processor has a clock rate of 2.5 GHz and a CPI of 1.0 and P3 processor has 4.0 GHz clock rate and has a CPI of 2.2.
- (a) Which processor has the highest performance expressed in instructions per second?
5. If the processors each execute a program in 10 seconds, find the number of cycles and the number of instructions? 10
- (b)
- (c) Let's consider a situation, where the execution time is reduced by 30%, which result an increase of 20% in the CPI. What clock rate should we have to get this time reduction?

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Winter Semester 2023-2024 Continuous Assessment Test – I

Program Name & Branch: BCB, BCE, BCI, BCT, BDS, BKT

SLOT: D2+TD2

Course Name & code: Computer Architecture and Organization- BCSE205L

Class Number (s): Common to all

Exam Duration: 90 Min.

Maximum Marks: 50

General instruction(s):

Answer all five questions

Q.No.	Question	Max Marks
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2.	a. Draw the flowchart for Booth's algorithm. b. Perform Booth's multiplication for the given numbers -15 and 6 (i.e. -15×6). Description of each step is expected.	10 [4+6]
3.	a. Explain the IEEE 754 floating-point representation. b. Perform the division for the given numbers 21 and 4 (i.e. $21/4$) using non-restoring algorithm. Description of each step is expected.	10 [4+6]
4.	a. Discuss the instruction types of IAS machine with suitable example? b. Describe the functional components of a computer with a neat diagram?	10 [5+5]
5.	Write the assembly code for the given arithmetic statement. $A = (B+C) * D$ using 0,1,2 and 3 address instructions and compute memory traffic for 0,1,2 and 3 address machines. Assuming the following fields: addresses - 2 bytes, data values - 2 bytes, opcode - 8-bits and word length - 1 byte.	10 [5+5]

Q.No.	Question	Max Marks									
1.	Scheme of evaluation: <table border="1"> <thead> <tr> <th>Sl. No</th><th>Topic</th><th>Marks</th></tr> </thead> <tbody> <tr> <td>1.</td><td>Assembly Code</td><td>4</td></tr> <tr> <td>1.</td><td>Register flow of operations</td><td>6</td></tr> </tbody> </table> Write an Assembly language programming for the following expression using the IAS computer Instruction set and interpret the flow of the IAS computer (Register flow of operations). $R = P - Q$ Assume the memory locations 800, 801, and 802 for P, Q, and R respectively. Ans: $R = P - Q$ Assembly Code: 1. LOAD M[800] : $AC \leftarrow M[800]$ 2. SUB M[801] : $AC \leftarrow AC - M[801]$ 3. STORE M[802] : $M[802] \leftarrow AC$ 4. HALT <u>Register Flow of Operations</u>	Sl. No	Topic	Marks	1.	Assembly Code	4	1.	Register flow of operations	6	10
Sl. No	Topic	Marks									
1.	Assembly Code	4									
1.	Register flow of operations	6									

$MAR \leftarrow PC$ $MBR \leftarrow M[MAR]$ $IR \leftarrow MBR<0..39>$ $MAR \leftarrow MBR<8..19>$ $MBR \leftarrow M[MAR]$ $AC \leftarrow MBR$ $IR \leftarrow IR<0..7>$ $MAR \leftarrow IR<8..19>$ $MBR \leftarrow M[MAR]$ $AC \leftarrow AC - MBR$ 1 2	$PC \leftarrow PC+1$ $MAR \leftarrow PC$ $MBR \leftarrow M[MAR]$ $IR \leftarrow MBR<0..39>$ $MAR \leftarrow MBR<8..19>$ $MBR \leftarrow MBR<0..7>$ $MBR \leftarrow AC$ $M[MAR] \leftarrow MBR$ $IR \leftarrow IR<0..7>$ 3 4	
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2.	Scheme of evaluation: <table> <tr> <th>Sl. No</th><th>Topic</th><th>Marks</th></tr> <tr> <td>2.a</td><td>Booth recoded table</td><td>3</td></tr> <tr> <td>2.b</td><td>Steps</td><td>4</td></tr> <tr> <td>2.b</td><td>Description of each step</td><td>2</td></tr> <tr> <td>2.b</td><td>Final result</td><td>1</td></tr> </table>	Sl. No	Topic	Marks	2.a	Booth recoded table	3	2.b	Steps	4	2.b	Description of each step	2	2.b	Final result	1	10 [3 + 7]
Sl. No	Topic	Marks															
2.a	Booth recoded table	3															
2.b	Steps	4															
2.b	Description of each step	2															
2.b	Final result	1															

a. Show the Bit pair recoded table for three-bit combinations. 3 Marks.

Multiplier bit pair		Multiplier bit on the right	Multiplicand selected at position i
i+1	i	i-1	
0	0	0	$0 \times M$
0	0	1	$+1 \times M$
0	1	0	$+1 \times M$
0	1	1	$+2 \times M$
1	0	0	$-2 \times M$
1	0	1	$-1 \times M$
1	1	0	$-1 \times M$
1	1	1	$0 \times M$

b. Perform the multiplication with the numbers -12 and 6 using the Modified Booth algorithm. A description of each step is required. (7 Marks)

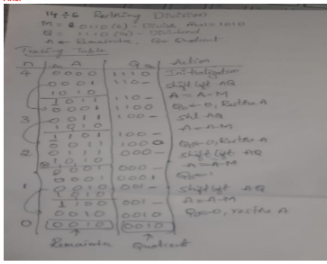
Ans:

3.	Scheme of evaluation:		10
	Sl. No	Topic	Marks
	3.a	Identification	1
	3.a	Justification	3
	3.b	Steps	4
	3.b	Description of each step	2
	3.b	Final result	1

a. Identify the most widely used signed number representation among the available representations. Justify your answer. 4 Marks

Ans: 2's Complement Representation is most widely used for signed number representation.
 Other representations and their limitations are as follows.

- Signed magnitude representation:
 - 2 representations for 0
 - Simple
 - 255 different numbers can be represented.
 - Need to consider both sign and magnitude in arithmetic
 - Different logic for addition and subtraction
- 1's complement representation:
 - 2 representations for 0
 - Complexity in performing addition and subtraction
 - 255 different numbers can be represented.
- 2's complement representation:
 - Only one representation for 0
 - 256 different numbers can be represented.
 - Arithmetic works easily

	b. Perform the division with the numbers 14 and 6 (ie. 14/6) using the restoring algorithm. A description of each step is required. (7 Marks)		10
	Ans: 		[5 + 5]
4.	Scheme of evaluation:		10
	Sl. No	Topic	Marks
	4.a	RISC vs CISC	5
	4.b(i)	Binary Conversion	1
	4.b(ii)	Arithmetic Shift Right	1.5
	4.b(iii)	Binary Conversion	1
	4.b(iv)	Arithmetic Shift Right	1.5

a. Compare and Contrast RISC with CISC. (5 Marks)

Ans:

CISC	RISC
Emphasis on hardware	Emphasis on software
Includes multi-lookup	Simple-lookup

Q. No.	Question	Max Marks									
1.	a) Draw the logic diagram of a register file with three access ports and explain the operation of $R2 \leftarrow R1 + R2$ using register files. Key: Scheme of Evaluation <table border="1"> <thead> <tr> <th>S.No</th> <th>Sub Topic</th> <th>Marks</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>Logic Diagram</td> <td>3</td> </tr> <tr> <td>2</td> <td>Explanation</td> <td>2</td> </tr> </tbody> </table> Logic Diagram: Ex: $R2 \leftarrow R1 + R2$ Read Address A = 01 Read Address B = 10 Write Address C = 10 • Explanation must be given	S.No	Sub Topic	Marks	1	Logic Diagram	3	2	Explanation	2	5
S.No	Sub Topic	Marks									
1	Logic Diagram	3									
2	Explanation	2									

b) Compare and contrast RISC and CISC architectures. Also, Perform $C = A + B$ using RISC and CISC instructions.	5
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Scheme of Evaluation

S.No	Sub Topic	Marks
1	Comparison	3
2	Code and explanation	2

Key:

Characteristics of RISC

1. It has simpler instructions and thus simple instruction decoding.
2. More general-purpose registers.
3. The instruction takes one clock cycle in order to get executed.
4. The instruction comes under the size of a single word.
5. Few data types.
6. Simpler addressing modes.
7. Pipeline can be easily achieved.

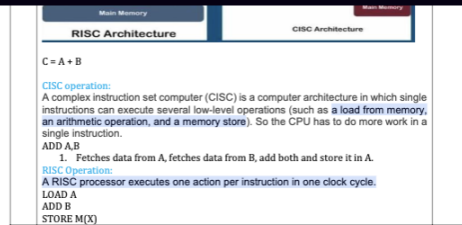
Characteristics of CISC

1. Instructions are complex, and thus it has complex instruction decoding.
2. Lesser general-purpose registers since the operations get performed only in the memory.
3. The instructions may take more than one clock cycle in order to get executed.
4. The instruction is larger than one-word size.
5. More data types.
6. Complex addressing modes.
7. CISC is not pipelined or less pipeline.

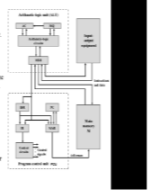
RISC VS CISC

RISC Architecture

CISC Architecture



	- The Load instruction stores the data from a memory location A into accumulator. - Add operation performs addition on the content of B with accumulator and stores the result in accumulator. - Finally, store will transfer the accumulator content to the memory location A.													
2.	Explain the various registers of IAS machine and write the significance of each register. Also demonstrate an assembly code and Register Transfer operation using IAS instructions for the following expression: $S = ((U+V)W)/X$ Assume the inputs are available in the memory locations 201 onwards. Store the result in 206 onwards. Scheme of Evaluation <table border="1"> <thead> <tr> <th>S.No</th><th>Sub Topic</th><th>Marks</th></tr> </thead> <tbody> <tr> <td>1</td><td>Significance of IAS registers</td><td>3</td></tr> <tr> <td>2</td><td>Assembly code</td><td>3</td></tr> <tr> <td>3</td><td>Register Transfer operation</td><td>4</td></tr> </tbody> </table> Solution: Set of registers (storage in CPU) Memory Buffer Register (MBR) - Contains a word to be stored in memory or it is used to receive a word from memory or from the I/O unit. Memory Address Register (MAR) - Specifies the address in memory of the word to be written from or read into the MBR. Instruction Register (IR) - Contains the 8 bit opcode instruction being executed. Instruction Buffer Register (IBR) - Employed to hold temporarily the right hand instruction from a word in memory. Program Counter (PC) - Contains the address of the next instruction pair to be fetched from memory. Accumulator (AC) & Multiplier Quotient (MQ) - Employed to hold temporarily the right hand instruction from a word in memory. For eg. The result of multiplying two 40 bit numbers is an 80 bit number, the most significant 40 bits are stored in the AC and the least significant in the MQ. Assembly code:	S.No	Sub Topic	Marks	1	Significance of IAS registers	3	2	Assembly code	3	3	Register Transfer operation	4	10
S.No	Sub Topic	Marks												
1	Significance of IAS registers	3												
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Q. No	Question	M	CO	BL												
1.	Scheme of Evaluation <table> <tr> <th>Sl. No.</th><th>Topic</th><th>Marks</th></tr> <tr> <td>1</td><td>List of registers (6)</td><td>3</td></tr> <tr> <td>2</td><td>Brief Description on each</td><td>4</td></tr> <tr> <td>3</td><td>RISC Vs CISC (Min. 3 comparisons for each)</td><td>3</td></tr> </table> a. Explain the importance of different registers in the IAS computer and illustrate their interconnections with a neat block diagram. Von Neumann Machine MBR: Memory Buffer Register - contains the word to be stored in memory or just received from memory. MAR: Memory Address Register - specifies the address in memory of the word to be stored or retrieved. IR: Instruction Register - contains the instruction currently being executed. IBR: Instruction Buffer Register - temporary store for IR's instruction (also used as memory). PC: Program Counter - address of next instruction to be fetched from memory. ALU: Accumulator & MQ: Multiplier-Quotient - holds operands and results of ALU ops.  b. Compare CISC and RISC architecture.	Sl. No.	Topic	Marks	1	List of registers (6)	3	2	Brief Description on each	4	3	RISC Vs CISC (Min. 3 comparisons for each)	3	7	CO1	1
Sl. No.	Topic	Marks														
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Page 1 of 6



REG.NO.:

NAME OF THE SCHOOL
CONTINUOUS ASSESSMENT TEST - I
WINTER SEMESTER 2024-2025

SLOT: C1 + TC1

RISC	CISC									
It is a Reduced Instruction Set Computer	It is a Complex Instruction Set Computer									
It focuses on Software.	It focuses on Hardware.									
It uses only a Hardwired control unit.	It uses both hardwired and microprogrammed control units.									
Translators are used for more registers	Translators are used for storing complex instructions.									
Code size is large.	Code size is small.									
The users of the pipeline are simple in RISC.	Users of the pipeline are difficult in CISC.									
An instruction is executed in a single clock cycle.	Instruction may take more than one clock cycle.									
An instruction can fit in one word.	Instructions are larger than the size of one word.									
The execution time of RISC is very short.	The execution time of CISC is longer.									
The program written for RISC architecture needs to take more space in memory.	The Program written for CISC architecture needs to take less space in memory.									
2.	CO1 1									
<table><tr><th colspan="2">Scheme of Evaluation</th><th rowspan="3">Marks</th></tr><tr><th>Sl. No.</th><th>Topic</th></tr><tr><td>1</td><td>Program</td></tr><tr><td></td><td>4</td></tr></table>		Scheme of Evaluation		Marks	Sl. No.	Topic	1	Program		4
Scheme of Evaluation		Marks								
Sl. No.	Topic									
1	Program									
	4									

Scheme of Evaluation		
Sl. No.	Topic	Marks
1	Program	4
2	Description	3
3	Memory Layout	3

Write an assembly code to execute the following operation and interpret to Von Neumann architecture

$$Z = ((A+B) + (C-D)) * E$$

Note: Program can be stored at memory location 300 onwards and Data variables are available from location address 500 onwards.
Assembly Language : [5 marks]

1	LOAD M(500)	Acc ← M[500]
2	ADD M(501)	Acc ← Acc + M[501]
3	STORE M(505)	M[505] ← Acc
4	LOAD M(502)	Acc ← M[502]
5	SUB M(503)	Acc ← Acc - M[503]
6	STORE M(506)	M[506] ← Acc
7	LOAD M(505)	Acc ← M[505]

Page 2 of 6



REG.NO.:

SLOT: C1 + TC1

8	ADD M(506)	Acc ← Acc + M[507]
9	STORE M(505)	M[508] ← Acc
10	LOAD MQ,M(505) MQ ← M[508]	
11	MUL M(504)	Acc, MQ ← M[504] x MQ
12	STORE M(505)	M[509] ← Acc
13	LOAD MQ	Acc ← MQ
14	STORE M(506)	M[510] ← Acc

Memory

Layout		
Instructions	300	1 2
	301	3 4
	302	5 6
	303	7 8
	304	9 10
	305	11 12
	306	13 14
	---	---
	---	---
	---	---
Data	500	A
	501	B
	502	C
	503	D
	504	E
	505	Z
	506	Z+1

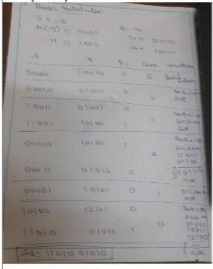
3.				C01	3												
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		10															

Page 3 of 6



REG.NO.:

SLOT: C1 + TC1

Calculate 13×14 using Booth's multiplication algorithm with explanation of the stepwise operation 																
4.			C01	3												
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Scheme of Evaluation																
Sl. No.	Topic	Marks														
1	Table of steps	6														
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		6														
a) Perform non-restoring division with a dividend of 21 and a divisor of 4.																

Page 4 of 6